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Promoting Accurate Variability of Social Skills in Children With Autism Spectrum Disorder

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Abstract

Restricted and repetitive behavior is a central feature of autism spectrum disorder (ASD), with such behaviors often resulting in lack of reinforcement in social contexts. The present study investigated training multiple exemplars of target behaviors and the utilization of lag schedules of reinforcement in the context of social skills training to promote appropriate and varied social behavior in children with ASD. Five participants with ASD between the ages of 7 and 9 attended a twice-weekly social skills group for 8 weeks. A multiple probe design across skills was utilized to assess intervention effects. During baseline, participants demonstrated low levels of skill accuracy and low appropriate variability in responding. During continuous reinforcement with one trained exemplar, skill accuracy increased while appropriate variability remained low. Training of three exemplars of target skills resulted in minimal improvements in appropriate variability. Introduction of a Lag 2 schedule with three trained exemplars was generally associated with increased appropriate variability. Further appropriate variability was observed during Lag 4 with three trained exemplars. Limitations and implications are discussed.

Keywords

social skills, behavioral variability, lag schedules, autism spectrum disorder

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Individuals with autism spectrum disorder (ASD) demonstrate social communication impairments and restricted and repetitive behaviors (American Psychiatric Association, 2013). Although they represent two distinct diagnostic criteria, these features of ASD often co-manifest in social settings. Individuals with ASD often demonstrate a narrow range of conversational topics (Lee, McComas, & Jawor, 2002), rote conversational exchanges (Leekam, Prior, & Uljarevic, 2011), and restricted responding to social initiations of others (Wolfe, Slocum, & Kunnavatana, 2014). Although lack of variability in responding may be functional in some contexts (e.g., completing math problems), invariability in social settings may be aversive to peers, resulting in an inability to adapt effectively in the presence of environmental changes and hindering problem solving (e.g., Mercier, Mottron, & Belleville, 2000; Parsonson & Baer, 1978; Wolfe et al., 2014).

Several factors may contribute to invariability in social behaviors, such as a limited behavioral repertoire (Lee, Sturmey, & Fields, 2007; Wolfe et al., 2014). To address invariability which results from a restricted repertoire, it has been suggested that training multiple exemplars may be effective (Goldsmith & Leblanc, 2004; Sherer et al., 2001); however, research in utilizing multiple exemplars to increase appropriate behavioral variability in individuals with ASD has been mixed. For example, Betz, Higbee, Kelley, Sellers, and Pollard (2011) provided reinforcement for a mand frame the first time it was demonstrated, followed by extinction for subsequent demonstrations of the same mand frame. Results indicated that this procedure was insufficient for increasing mand variability in children with ASD—with variability only increasing following training of three distinct mand frames via script fading. Other researchers have found training of multiple exemplars of mands, either in isolation or in combination with a differential reinforcement procedure, to be insufficient to increase appropriate behavioral variability (e.g., Sellers, 2011).

Alternatively, invariability in social behaviors may be a function of an environment that does not require or support varied responding (Lee et al., 2007). In other words, an environment in which reinforcement is provided irrespective of response variability is unlikely to elicit appropriate and variable responding. An example of such an environment may be a teacher or parent providing reinforcement each time a child appropriately responds to the question, "how are you today," even if the child always responds with "fine." Basic research has suggested that dense schedules of reinforcement, such as continuous schedules of reinforcement, often suppress appropriate behavioral variability (e.g., Vogel & Annau, 1973). Similarly, fixed-ratio (FR) schedules that result in frequent reinforcement (e.g., FR 1-FR 300) have been found to result in restricted behavioral variability (Boren, Moerschbaecher, & Whyte,

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1978). Findings of schedule-restricted variability have also been replicated in individuals with ASD. For example, Heldt and Schlinger (2012) found low levels of tacting variability when praise and preferred stimuli were provided following each correct tact regardless of whether appropriate variability was demonstrated.

Although restricted and repetitive social behaviors are likely to result in poor outcomes (e.g., Mercier et al., 2000), few social skills interventions systematically plan and assess for appropriate variability promoted via training of multiple exemplars or manipulation of schedules of reinforcement. Whereas incorporation of multiple exemplars of target skills is often utilized to promote generalization of target social skills (e.g., Plavnick, Kaid, & MacFarland, 2015), many interventions utilize dense FR schedules of reinforcement with no modification of reinforcement schedule that would promote appropriate variability (e.g., Jones, Lerman, & Lechago, 2014; Mason et al., 2014). Given that the FR schedules of reinforcement frequently utilized in social skills training may serve to reinforce social behavior stereotypy (Boren et al., 1978), other schedules of reinforcement should be considered when addressing restricted and repetitive social behavior in individuals with ASD.

A lag schedule is a schedule of reinforcement in which reinforcement is delivered following a response that differs from a prespecified number of previous responses (Page & Neuringer, 1985). Using a Lag 2 schedule, for example, reinforcement would be provided following a response that differed from the two responses previously emitted. Lag schedules of reinforcement have been utilized to increase appropriate variability of tacting (Lag 3; Heldt & Schlinger, 2012), play behaviors (Lag 1 and Lag 2; Lang et al., 2014; Lag 1; Napolitano, Smith, Zarcone, Goodkin, & McAdam, 2010), social questions (Lag 1; Lee et al., 2002; Lag 1; Lee & Sturmey, 2006), and conversations (Lag 1; Lee & Sturmey, 2014), suggesting that lag schedules of reinforcement may be utilized to increase the appropriate variability of behaviors frequently targeted in social skills curriculum. In a review of lag schedule use targeting restricted and repetitive behaviors in children with ASD, Wolfe et al. (2014) found that the majority of participants demonstrated 3 to 6 times more variability under lag schedule conditions than other non-lag schedules—even though most studies utilized a relatively small schedule (i.e., Lag 1). Although relatively few studies have assessed maintenance of appropriate variability, those that have reported maintained appropriate variability (Heldt & Schlinger, 2012; Lang et al., 2014; Napolitano et al., 2010).

In general, researchers have found that teaching new responses and introducing schedules of reinforcement to promote varied responding are useful in addressing restricted and repetitive social behaviors in individuals with ASD

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(e.g., Wolfe et al., 2014). Although research suggests that training multiple exemplars of target skill use and incorporation of lag schedules may be effective in addressing restricted and repetitive behaviors, instructional strategies such as these are often underutilized unless incorporated into practice-ready curricula (e.g., Dingfelder & Mandell, 2011; Kasari & Smith, 2013). Although no existing curricula incorporate lag schedules of reinforcement as a strategy for promoting appropriate variability of social behaviors, several have incorporated training of multiple exemplars as a strategy for promoting generalization of target skills (Stokes & Osnes, 1989).

The Superheroes Social Skills program (Jenson et al., 2011) is an example of a social skills curriculum that incorporates training of multiple exemplars of behavior (e.g., diverse video models depicting variations of each target skill; diverse role-plays) as a strategy to promote accurate demonstration and generalization of target skills. The program, comprised of 18 lessons targeting different social skills, utilizes animated superheroes on DVD to teach target skills. The Superheroes Social Skills program also incorporates several other empirically supported strategies to promote skill acquisition: behavioral skills training (e.g., Stewart, Carr, & LeBlanc, 2007), in which participants role-play and receive performance feedback regarding skill use; video models depicting three to five examples of skill use by unknown peers (e.g., Buggey, Hoomes, Sherberger, & Williams, 2011); self-monitoring of successful skill use (Lee, Simpson, & Shogren, 2007); and social stories (e.g., Scattone, Tingstrom, & Wilczynski, 2006), presented in the form of comic books.

Purpose of the Study

Prior evaluations of the Superheroes Social Skills curriculum have found improvements in skill accuracy following implementation of training (Radley et al., 2015; Radley, O'Handley, et al., 2014); however, this research has not determined whether implementation of the intervention, which incorporates multiple exemplars of target social skills, is effective in addressing restricted and repetitive social behaviors. Although some research suggests that training multiple exemplars of behavior may be effective in addressing restricted and repetitive behaviors when combined with reinforcement and extinction procedures (Betz et al., 2011), other research suggests similar strategies to be ineffective (Sellers, 2011). Lag schedules of reinforcement have been found to be effective in promoting appropriate variability of behaviors (e.g., Lang et al., 2014; Lee & Sturmey, 2014; Napolitano et al., 2010), but are not widely adopted into social skills curricula. Despite the promise of lag schedules in addressing restricted and repetitive behaviors, some have suggested that the

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relatively small lag requirements utilized in most studies (e.g., Lag 1; Wolfe et al., 2014) may result in higher order stereotypy—in which individuals develop patterned responding (Lee et al., 2002). Some have suggested that larger lag schedules may effectively address patterned responding (Rodriguez & Thompson, 2015), with other research documenting that larger lag schedules may provide little additional benefit in terms of appropriate variability (Baruni, Rapp, Lipe, & Novotny, 2014). Last, lag schedule research has been limited in that few studies have determined whether changes in appropriate behavioral variability are maintained following removal of the lag schedule of reinforcement.

Given the limitations of previous research on the Superheroes Social Skills program and on lag schedules of reinforcement, the present study sought to evaluate the effect of participation in the Superheroes Social Skills program as typically delivered and when modified to incorporate lag schedules of reinforcement on accuracy and appropriate variability of target social skills. The Superheroes Social Skills program was selected as the curriculum to be used in the current study due to the fact that it incorporates multiple exemplars of training but lacks programming for social skill variability. As such, researchers were able to assess the effect of training multiple exemplars on social skill variability in isolation and following the sequential introduction of lag schedules of reinforcement. The present study also sought to extend the literature through evaluation of the use of larger lag schedules for promoting additional appropriate variability, and determination of maintenance of appropriate variability following termination of lag schedules of reinforcement. The following research questions guided the present study:

Research Question 1: Does training participants in multiple response exemplars result in increased skill accuracy and appropriate variability?
Research Question 2: Does the addition of lag schedules of reinforcement result in greater appropriate variability in target skills than training utilizing multiple response exemplars alone?
Research Question 3: Does increasing the lag schedule response requirement beyond the number of trained exemplars result in response generalization (i.e., demonstration of untrained, appropriate and variable responses)?

Method

Participants and Settings

Participants in the current study included five children with ASD who were referred to a university-based clinic for social skills training. Inclusionary

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criteria were a diagnosis of ASD, between the ages of 7 and 10, and parent report of social skill deficits. All participants had previously received diagnoses of ASD from licensed psychologists and had special education rulings of autism from school-based multidisciplinary teams. Mark was a 9-year 6-month–old Caucasian male in the third grade. Mark received special education services at school in a self-contained classroom with additional speech and language services. Jarvis was an 8-year 3-month–old African American male in the third grade. He received pull-out special education services to address academic skills and behavior. Phillip was a 7-year 7-month–old Caucasian male in the second grade. Phillip received special education services addressing speech and language skills. Jarrod was a 7-year 6-month–old African American male in the first grade. Jarrod received special education services to address academic skills and speech and language skills. Tony was a 7-year 2-month–old Caucasian male in the first grade. Tony received special education services to address disruptive behavior.

The five participants attended a social skills group with one typically developing, same-age (i.e., 8-year 4-month) peer. The social skills group and assessment of skills accuracy and appropriate variability took place at a university-based clinic in the southeastern United States in a conference room. The room had an oval table and chairs placed in the center of the room. The room also contained a dry erase board and 40-inch television used to display training videos.

Dependent Measures

Appropriate variability. The primary dependent variable was the number of consecutive correct non-repeated responses provided by participants in response to research assistant prompts related to four social skills. The four target skills were selected from the Superheroes Social Skills program based on parent-reported social skill deficits that were common across all participants with ASD. The target skills selected from the Superheroes Social Skills program were Responding to Questions, Maintaining a Conversation, Expressing Wants and Needs, and Participate (see Table 1). To assess appropriate variability for each of the target skills, researchers provided participants with a verbal prompt to elicit a target response (i.e., “What do you like to do for fun,” “Tell me about your day,” “Go ask [name] for a toy,” and “Go play with [name]”). Researchers recorded verbatim the physical (if applicable) and verbal response demonstrated by the participant (e.g., tapped on shoulder and said “can I play with you?”).

A correct variable response was defined as a response that was topographically different from the previous responses but was contextually appropriate

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Table 1. Task Analyses of Target Skills.

Responding to questions	Maintaining a conversation	Expressing wants and needs	Participation
1. Face the person with head and shoulders oriented toward partner	1. Face the person with head and shoulders oriented toward partner	1. Face the person with head and shoulders oriented toward partner	1. Demonstrate close physical proximity to interaction partner/activity
2. Make and sustain eye contact for a minimum of 3 s	2. Make and sustain eye contact for a minimum of 3 s	2. Make and sustain eye contact for a minimum of 3 s	2. Face the person with head and shoulders oriented toward partner
3. Demonstrate listening by maintaining orientation, sustaining eye contact, and nodding head	3. Demonstrate listening by maintaining orientation, sustaining eye contact, and nodding head	3. Wait for a pause in the conversation without interrupting	3. Make and sustain eye contact for a minimum of 3 s
4. Provide an appropriate response to the question	4. Wait for turn without talking over or interrupting the speaker	4. Give a signal that you want to talk (e.g., raise hand, tap shoulder twice, say person's name)	4. Wait for turn without skipping others or interrupting turn-taking progression
	5. Make an appropriate comment or ask a question about what the person says	5. Wait for the person to respond to the signal	5. Join in without disrupting the progression of activity using an activity-appropriate response
		6. Say what is wanted/needed	

for the prompt provided. Similar responses that shared one or more words were considered correct if omission of the shared words still resulted in a contextually appropriate response. For example, the two verbalizations "I like to play with friends" and "I like to go to the park" were both considered correct variable responses in reply to "What do you like to do for fun?" because

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omission of the shared words "I like to" left two topographically different and contextually appropriate responses: "Go to the park" and "Play with friends." Furthermore, responses that had the same meaning as a previous response and consisted only of a portion of a previous response were considered repetitions. Therefore, when presented with a cue for Maintaining a Conversation, "My day was great" and "Great" would be considered repetitions.

Each of the four skills examined in this study were comprised of multiple steps, many of which were identical across skills (e.g., face the person with head and shoulders oriented toward partner, make and sustain eye contact for a minimum of 3 s); however, variable responding within each skill was evaluated based on the participants' performance of a single step that was unique to each skill. In regard to Responding to Questions, the researchers evaluated appropriate variability of verbal responses during the fourth step (i.e., provide an appropriate response to the question) of the skill. Following the Maintaining a Conversation cue, the appropriate variability of verbal responses were evaluated within the fifth step (i.e., make an appropriate comment or ask a question about what the person says). Variability for Expressing Wants and Needs was examined during the sixth step (i.e., say what is wanted/needed). Appropriate variability was evaluated based on request frames. For instance, when prompted to "Go ask [name] for a toy," the response "May I have that dinosaur?" would be a varied response than "Can I play with that dinosaur?" In other words, although the mand is the same across both responses (i.e., "dinosaur"), the differences across the request frames (i.e., "May I have" and "Can I play with") were considered sufficient for the purpose of determining appropriate variability. In addition, different mands in identical frames were considered variable responses so that "May I have the dinosaur?" was considered to vary from "May I have the puzzle?" Finally, in the Participation skill (i.e., join in without disrupting the progression of activity using an activity-appropriate response), the researchers evaluated the appropriate variability of the fifth step (i.e., join in using an activity-appropriate response). For example, when provided the cue to "Go play with [name]," the response of saying the person's name was novel compared with tapping that person on the shoulder.

Skill accuracy. A secondary dependent variable in the present study was skill accuracy. Skill accuracy was assessed simultaneously with appropriate variability. Following delivery of a prompt to demonstrate the target skill, researchers coded the participants' accuracy of demonstrating the target skill using the task analysis derived from the Superheroes Social Skills manual (Table 1). Accuracy was defined as the percentage of steps in the task analysis that were correctly demonstrated by the participant for a particular target

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skill. It should be noted that accuracy in this sense was measured separately from context appropriateness of the verbalization, which was measured as part of appropriate variability.

Social validity. Following the completion of social skills training, the parents of the participants were asked to complete the Behavior Intervention Rating Scale (BIRS; Elliott & Treuting, 1991). The questionnaire contains 24 Likert-type scale items to assess the parents' perception of the acceptability, effectiveness, and time of effectiveness of the social skills training program. The psychometrics measured for the BIRS was internal consistency with alpha values for the total being .97 and alpha values of .97, .92, and .87 for the Acceptability, Effectiveness, and Time of Effectiveness factors, respectively (Elliott & Treuting, 1991).

Design

A multiple probe design across skills with concurrent replication across participants was used to evaluate the effects of social skills training on appropriate variability and skill accuracy. Within a multiple probe design, data are collected intermittently—allowing for the collection of data on a larger number of subjects or skills than may otherwise be possible while resulting in data that differ minimally from frequent observation (Bijou, Peterson, Harris, Allen, & Johnston, 1969). The study consisted of six experimental phases: baseline, behavioral skills training with one trained exemplar and continuous reinforcement, behavioral skills training with three trained exemplars and continuous reinforcement, behavioral skills training with three trained exemplars and a Lag 2 schedule of reinforcement, behavioral skills training with three trained exemplars and a Lag 4 schedule of reinforcement, and maintenance. Phase changes occurred based on visual analysis of appropriate variability data (i.e., stability of responding or increased appropriate variability in responding). Data were analyzed through the visual analysis of level, trend, variability, consistency across similar phases, overlap, and immediacy of effect (Kratochwill et al., 2010) for appropriate variability and skill accuracy.

Procedures

Baseline. Prior to the beginning of social skills training, the participants' parents completed the Autism Social Skills Profile (ASSP; Bellini & Hopf, 2007) in order for the researchers to identify skills to be targeted in training. Following the identification of target skills, research assistants provided the participants with the aforementioned cues to demonstrate the four target

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skills. No performance feedback was provided during this phase of the study. For the first skill to be trained (i.e., Responding to Questions), nine probes were administered during a single baseline session. For the remaining skills, four probes were collected concurrent with probing of Responding to Questions. Cues for target social skills were provided at the beginning of each probe. Probes were conducted in the order in which skills were to be taught.

Intervention. Participants attended a 2-hr social skills training group twice per week across 8 weeks. The Superheroes Social Skills program was utilized during all phases of the intervention using procedures identical to those described by Radley, O'Handley, et al. (2014) and Radley et al. (2015). At the beginning of the social skills training session, the daily schedule and group rules were reviewed with the participants. Next, the target social skill was introduced to the participants using an animated video of superhero characters who explained the rationale for using the skill and the steps needed to demonstrate the skill. Then, the participants watched two to three videos of peer models correctly demonstrating the use of the target skill in various situations. Subsequently, the participants were trained to accurately use the skill through behavioral skills training involving modeling, role-playing, and performance feedback. During behavioral skills training, participants were able to earn marks on a skill monitoring card for correct use of the skill. Behavioral skills training was repeated until participants demonstrated three consecutive examples of 100% skill accuracy. Following behavioral skills training, the participants watched a digital social story depicting the superhero characters using the target skill and played a social game focusing on the target skill. Finally, data collection probes were conducted by having research assistants interact with the participants to elicit target skills. During each session, a minimum of nine probes were collected for the skill being taught, dependent on data stability. As with baseline, cues for skill use were provided at the beginning of each probe. Data for the remaining skills were collected consistent with the multiple probe design, with one to three probes collected per session. Probes for skills still in baseline were collected in the order in which skills were to be taught, and cues for skills were provided at the beginning of each probe.

The intervention phase included four experimental conditions that varied based on the goal of the behavioral skills training. During the behavioral skills training with one trained exemplar and continuous reinforcement phase, only one possible response to the skill prompt was modeled for the participants. The participants were provided with verbal praise and marks on the skill monitoring card for every instance of appropriate skill use. If participants failed to demonstrate the skill accurately, performance feedback was

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provided and the previously modeled response was provided. If a participant earned at least eight marks on his skill monitoring card, he was provided with a tangible prize (i.e., various toys) at the end of the training session.

In the next phase, behavioral skills training included three trained exemplars and continuous reinforcement. Therefore, three possible responses to the skill prompt were modeled for the participants. During role-plays, participants were provided with praise and a mark on their skill monitoring card for every instance of appropriate demonstration of the target skill—regardless of whether appropriate variability was demonstrated. Similar to the previous phase, participants were provided with performance feedback and one of the three previously modeled responses were provided for inaccurate skill use. In this phase, a participant was required to earn at least eight marks on his skill monitoring card to receive a tangible reward.

During the behavioral skills training with three trained exemplars and Lag 2 schedule of reinforcement phase, the same three response models that were presented in the previous phase were modeled prior to the participant engaging in role-plays. During behavioral skills training, participants were informed that marks on the self-monitoring card would only be provided following demonstration of three topographically different responses. When inaccurate skill use occurred, performance feedback was provided. If participants demonstrated a response that was not topographically different than either of the two preceding responses, they were provided with the prompt "You have said X and Y, try a different response next time." If they again failed to demonstrate a topographically different response, they were provided with the prompt "You have said X and Y, next time try saying Z" with Z representing one of the previously modeled responses. To account for the increased number of correct responses necessary to earn a mark on their skill monitoring card (i.e., 3), participants were provided with a tangible prize following the end of the training session if they had at least three marks on their cards.

In the final phase of the intervention, behavioral skills training included three trained exemplars and a Lag 4 reinforcement schedule. The same three responses that were modeled in the previous two phases were modeled during this phase as well. Similar to the previous phase, participants were informed during behavioral skills training that marks on the self-monitoring card would be provided for demonstrating appropriate and variable responding. Participants were told that marks on the self-monitoring card would be provided following demonstration of the fifth topographically different response. Similar to the previous phase, performance feedback was provided for incorrect skill use. In addition, the same prompts used in the previous phase for non-topographically different responses were provided when a participant provided a response that did not vary from any of the previous four responses.

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To account for the increased number of correct responses necessary to earn a mark on their skill monitoring card (i.e., 5), participants were provided with a tangible prize following the end of the training session if they had at least two marks on their cards.

Maintenance. Maintenance data were collected following mastery of a target skill. The maintenance phase consisted of providing participants with prompts for demonstrating the use of the target skills. Similar to the baseline phase, no reinforcement or performance feedback was provided for appropriate variability or skill accuracy.

Interobserver agreement (IOA). Trained graduate and undergraduate students independently recorded participants' verbatim responses to each prompt to calculate IOA for the topography and appropriate variability of responses. IOA for topography was calculated by dividing the number of agreements for each word or non-verbal behavior (e.g., tapping on a shoulder) in a response by the number of agreements and disagreements and multiplying by 100. IOA for appropriate variability was calculated by dividing the total number of agreements between raters regarding determination of appropriate variability of a response in comparison with prior responses (i.e., yes, response was variable; no, response was not variable) for all probes of a particular skill administered in one session by the number of agreements and disagreements for all probes of the same skill administered in one session and multiplying by 100. Finally, IOA was calculated for the participants correctly demonstrating each step of the target skill by dividing the number of agreements in completed steps by the number of agreements and disagreements and multiplying by 100. IOA was collected during a minimum of 20% of probes per phase. IOA was collected during 46.3% of observations for Mark with a mean of 98% (range = 66%-100%) for response topography, a mean of 100% for appropriate variability, and a mean of 97% (range = 60%-100%) for completed steps. For Jarvis, IOA was collected across 46% of observations with a mean of 95% (range = 83%-100%) for response topography, a mean of 100% for appropriate variability, and a mean of 96% (range = 40%-100%) for completion of steps. For Phillip, IOA was calculated for 40% of observations with a mean of 99% (range = 83-100%) for response topography, a mean of 100% for appropriate variability, and a mean of 95% (range = 40%-100%) for steps completed. IOA was calculated for 45% of observations for Jarrod with a mean of 98% (range = 66%-100%) for response topography, a mean of 100% for appropriate variability, and a mean of 99% (range = 80%-100%) for completion of steps. Finally, IOA was collected across 41% of observations for Tony with a mean of 98% (range = 83%-100%) for response topography, a

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mean of 100% for appropriate variability, and a mean of 98% (range = 50%-100%) for steps completed. When IOA fell below 80%, observers were retrained on the definitions of the target skills by the primary facilitator.

Treatment integrity. Treatment integrity was assessed during all intervention sessions. The primary facilitator completed a checklist derived from the Superheroes Social Skills manual and modified to assess both number of exemplars trained and implementation of lag schedule. Treatment integrity was calculated by dividing the number of steps completed by the number of possible steps and multiplying by 100. The mean treatment integrity during the intervention was 99%. An additional graduate student completed a treatment integrity checklist during 47% of intervention sessions. IOA for treatment integrity was calculated by dividing the number of agreements of steps completed by the number of agreements and disagreements and multiplying by 100. IOA for treatment integrity was 100%.

Results

Participant Outcomes

Mark. Figure 1 shows the results for appropriate variability and skill accuracy for all target skills for Mark. During baseline, Mark's responses were variable for Responding to Questions, providing a maximum of four novel responses; however, there was no appropriate variability for the Maintaining a Conversation, Expressing Wants and Needs, and Participate skills. Training one exemplar resulted in continued variable responding for the Responding to Questions skill, and a slight increase in consecutive correct non-repeated responses for Maintaining a Conversation. Although Mark provided a correct response during probes of Expressing Wants and Needs and Participate, no appropriate variability was observed. When trained with three exemplars, appropriate variability was not observed for Responding to Questions, Maintaining a Conversation, and Participate. A slight increase in the number of consecutive correct non-repeated responses was observed for Expressing Wants and Needs. Increased variable responding was observed following introduction of the Lag 2 phase for Responding to Questions, Maintaining a Conversation, Expressing Wants and Needs, and Participate. Increases in appropriate variability were observed for all skills during the Lag 4 phase, with the exception of Responding to Questions. Maintaining a Conversation was noted to reach particularly high levels of appropriate variability in relation to baseline previous phases. Finally, results indicate that Mark did not maintain variable responding for Responding to Questions, Maintaining Conversations, and Expressing Wants

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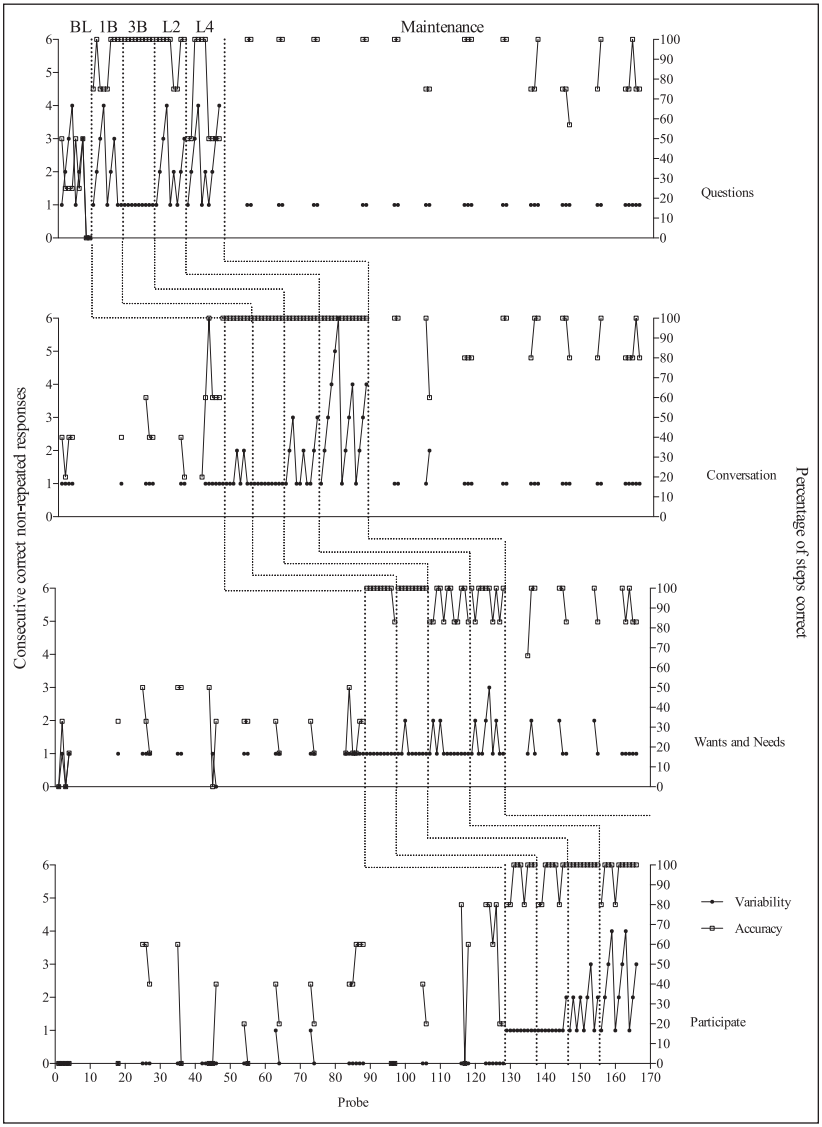


Figure 1. Consecutive correct non-repeated responses and percentage of steps correct, Mark.

and Needs following removal of lag schedule requirements, with data demonstrating clear overlap with data from the baseline phase.

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During baseline measures of skill accuracy, Mark demonstrated moderate levels of skill accuracy for all skills. An immediate increase in percentage of steps completed correctly was observed during the behavioral skills training with one exemplar (1B), with limited overlap between data points in experimental phases and baseline. Similar levels of skill accuracy were observed across subsequent experimental phases, with the exception of decrease in skill accuracy for Responding to Questions during the Lag 4 phase. Maintenance data for Mark indicated increased appropriate variability in skill accuracy; however, skill accuracy continued to demonstrate improvements over baseline levels with minimal overlap.

Jarvis. Figure 2 shows the results for response appropriate variability and skill accuracy for all target skills for Jarvis. Limited appropriate variability in correct responding was observed during baseline for Responding to Questions, Maintaining a Conversation, Expressing Wants and Needs, and Participate. No change in consecutive correct non-repeated responses was observed for Responding to Questions and Expressing Wants and Needs following training in one exemplar. Although Jarvis provided a correct response during Participate probes, minimal appropriate variability was observed. Increases in consecutive correct non-repeated responses were observed for the Maintaining a Conversation skill. However, complete overlap was observed with baseline data. Following training in three exemplars of the target skills, appropriate variability remained infrequent and at low levels, and no skills demonstrated non-overlap with baseline data. Immediately upon introduction of the Lag 2 phase, large increases in the number of consecutive correct non-repeated responses were observed for the Responding to Questions, Maintaining a Conversation, and Participate skills and non-overlap with data from previous phases was observed. Although magnitude of appropriate variability was small and non-overlap was not observed, increased appropriate variability was observed for Expressing Wants and Needs. With the exception of Expressing Wants and Needs, further increases in the number of consecutive correct non-repeated responses and non-overlap with data of previous phases were observed during the Lag 4 phase. Finally, removal of the lag schedule requirement resulted in appropriate variability similar to baseline phases.

During baseline measures of skill accuracy, Jacob demonstrated variable and moderate levels of skill accuracy across all target skills. Upon introduction of training with one exemplar, immediate increases in skill accuracy were observed. Similar levels of skill accuracy were observed across all intervention phases. Intervention phases were also associated with decreased appropriate variability and minimal overlap with baseline. Maintenance data

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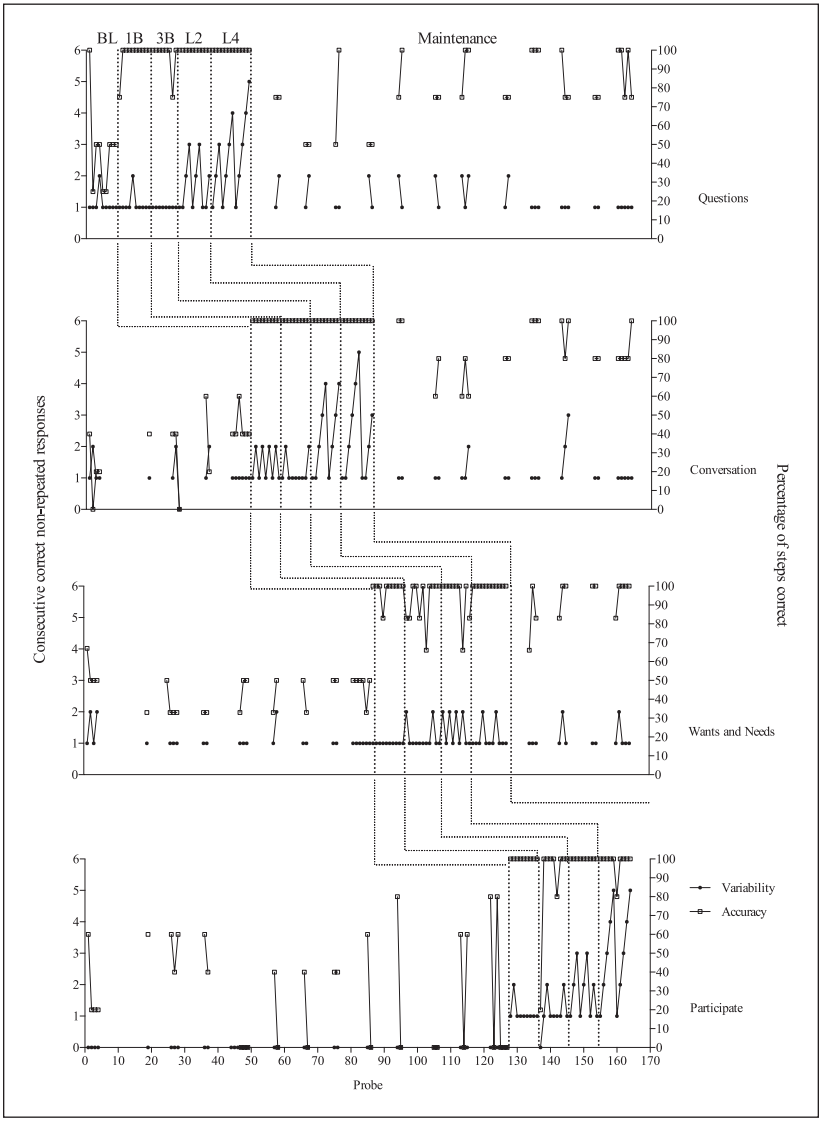


Figure 2. Consecutive correct non-repeated responses and percentage of steps correct, Jarvis.

indicated decreased skill accuracy and increased appropriate variability. However, mean levels remained improved over baseline.

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Phillip. Figure 3 shows the results for appropriate variability and skill accuracy for all target skills for Phillip. Little appropriate variability in responding was observed across skills during baseline probes. During training with one exemplar, a slight increase in consecutive correct non-repeated responses over baseline was observed for Maintaining a Conversation, Expressing Wants and Needs, and Participate, with Maintaining a Conversation and Participate demonstrating non-overlap with baseline. Due to absences, no data were collected during this phase for Responding to Questions. Similarly, absences precluded the collection of data during training with three exemplars for Maintaining a Conversation and Expressing Wants and Needs; however, training three exemplars was associated with slight increases in the frequency of appropriate variability, yet continued overlap, compared with baseline levels for Responding to Questions and Participate. Upon introduction of the Lag 2 phase, further increases in consecutive correct non-repeated responses were observed for Responding to Questions and Conversation. Non-overlap of data was observed for Responding to Questions. Similar levels of consecutive correct non-repeated responses were observed for Expressing Wants and Needs, and no data were collected for Participate due to absence. Across skills, the highest levels of consecutive correct non-repeated responses were observed during the Lag 4 phase, with improvements being observed for Responding to Questions and Maintaining a Conversation. Although improvements in the level of appropriate variability were smaller for Expressing Wants and Needs and Participate, all skills demonstrated non-overlap with data from previous phases. Across skills probed in maintenance, greater levels of consecutive correct non-repeated responses were observed compared with baseline levels. Non-overlap with baseline data was observed for Responding to Questions and Maintaining a Conversation.

During baseline measures of skill accuracy, variable and moderate to high levels of skill accuracy were observed across all skills. Immediate reductions in appropriate variability and increases in mean level were observed following the introduction of training in one exemplar. Similar high levels of skill accuracy were observed through the remaining experimental phases. Although results were variable, skill accuracy was generally maintained over baseline levels following removal of direct intervention procedures.

Jarrod. Figure 4 shows the results for response appropriate variability and skill accuracy and for all target skills for Jarrod. Baseline data indicate low levels of consecutive correct non-repeated responses across skills. A slight increase in consecutive correct non-repeated responses was observed for Responding to Questions and Maintaining a Conversation upon introduction of training with one exemplar, with no change in appropriate variability of responding for

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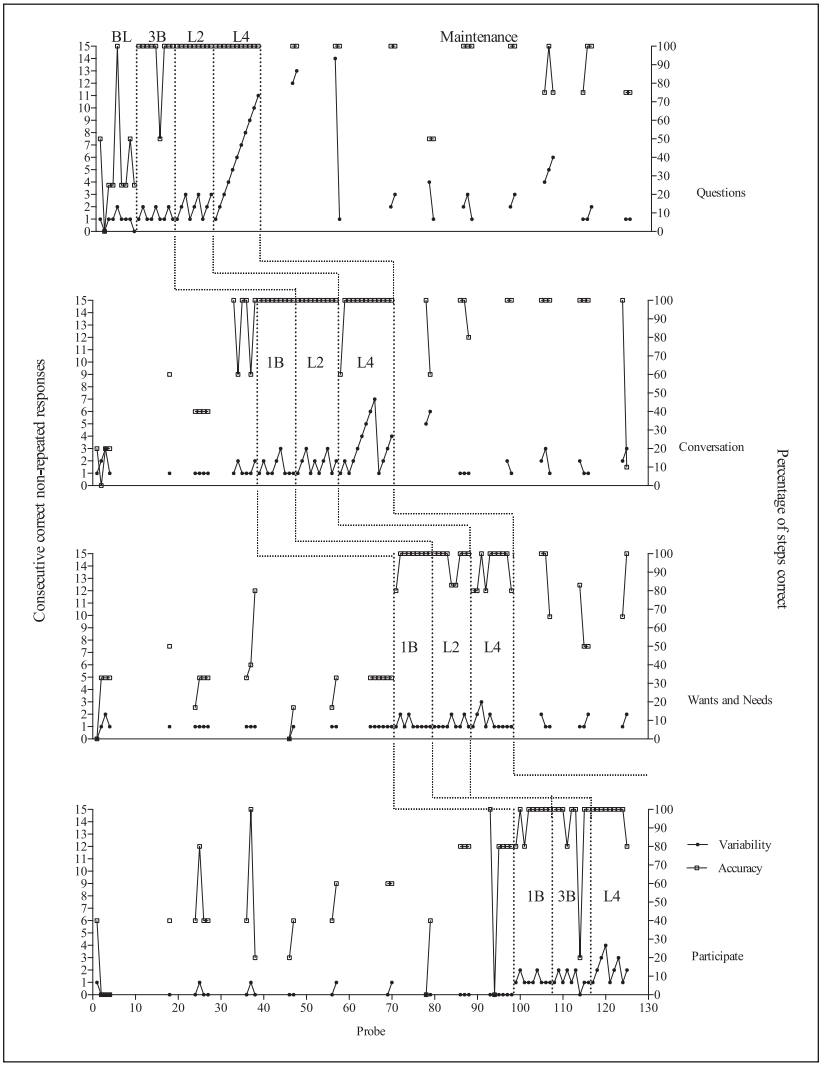


Figure 3. Consecutive correct non-repeated responses and percentage of steps correct, Phillip.

Expressing Wants and Needs. Non-overlap with baseline was only observed for Responding to Questions. During training with three exemplars, an increase in consecutive correct non-repeated responses and non-overlap with

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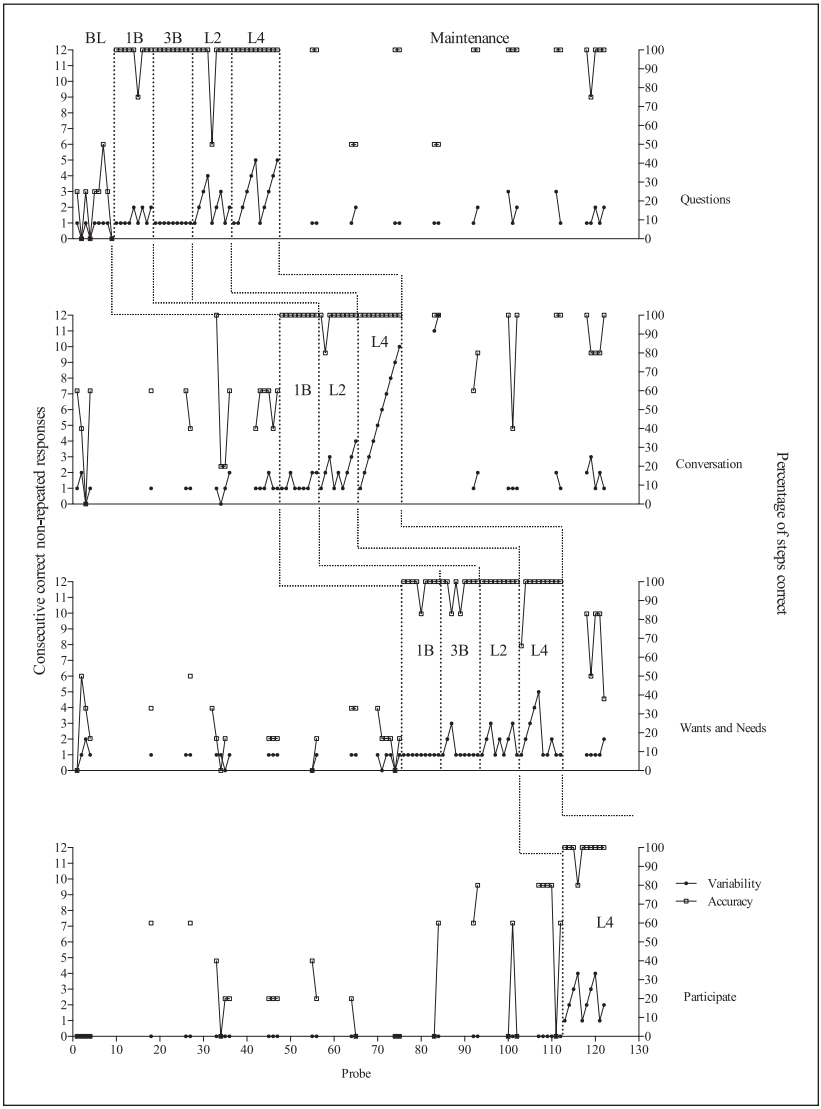


Figure 4. Consecutive correct non-repeated responses and percentage of steps correct, Jarrod.

previous phases was observed for Expressing Wants and Needs, whereas no appropriate variability was observed for Responding to Questions. Due to

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absence, no data were collected for Maintaining a Conversation. Upon introduction of the Lag 2 phase, immediate increases in consecutive correct non-repeated responses over all prior phases were observed for Responding to Questions, Maintaining a Conversation, and Expressing Wants and Needs. Both Responding to Questions and Maintaining a Conversation demonstrated non-overlap with previous phases. Further increases in variable responding were observed following introduction of Lag 4, with the highest levels of consecutive correct non-repeated responses observed during this phase. However, data for Expressing Wants and Needs demonstrated a decreasing trend during the final five probes in Lag 4. Due to absences, Jarrod was only exposed to Lag 4 for Participate. However, this phase demonstrated similar levels of consecutive correct non-repeated responses as other skills when in Lag 4. All skills demonstrated non-overlap with previous phases during Lag 4. Results indicated that appropriate variability only somewhat maintained for Responding to Questions and Maintaining a Conversation, but did not maintain for Expressing Wants and Needs. Compared with baseline levels, all skills probed in maintenance demonstrated small degrees of non-overlap.

During baseline measures, Jarrod demonstrated variable, low-to-moderate levels of skill accuracy. Upon introduction of training with one exemplar, immediate increases in mean level and reductions in appropriate variability were observed across skills. In addition, minimal levels of overlap were observed. Throughout all intervention phases, similar high levels of skill accuracy and minimal appropriate variability and overlap were observed for all skills. Results from the maintenance phase indicated decreased accuracy and increased appropriate variability for all skills. However, mean levels continued to remain above baseline levels.

Tony. Figure 5 shows the results for response appropriate variability and skill accuracy and for all target skills for Tony. Baseline results indicate little to no consecutive correct non-repeated responses across skills. A large increase in consecutive correct non-repeated responses was observed following training in one exemplar for Responding to Questions. However, all remaining skills demonstrated little to no response appropriate variability. Non-overlap with baseline was observed for Responding to Questions, Expressing Wants and Needs, and Participate. For Responding to Questions, a slight reduction in consecutive correct non-repeated responses was observed during training in three exemplars. A small increase in consecutive correct non-repeated responses was observed for both Maintaining a Conversation and Expressing Wants and Needs. Non-overlap with previous phases was only observed for Expressing Wants and Needs. No data were collected for Participate due to absence. Introduction of Lag 2 was associated with high levels of consecutive

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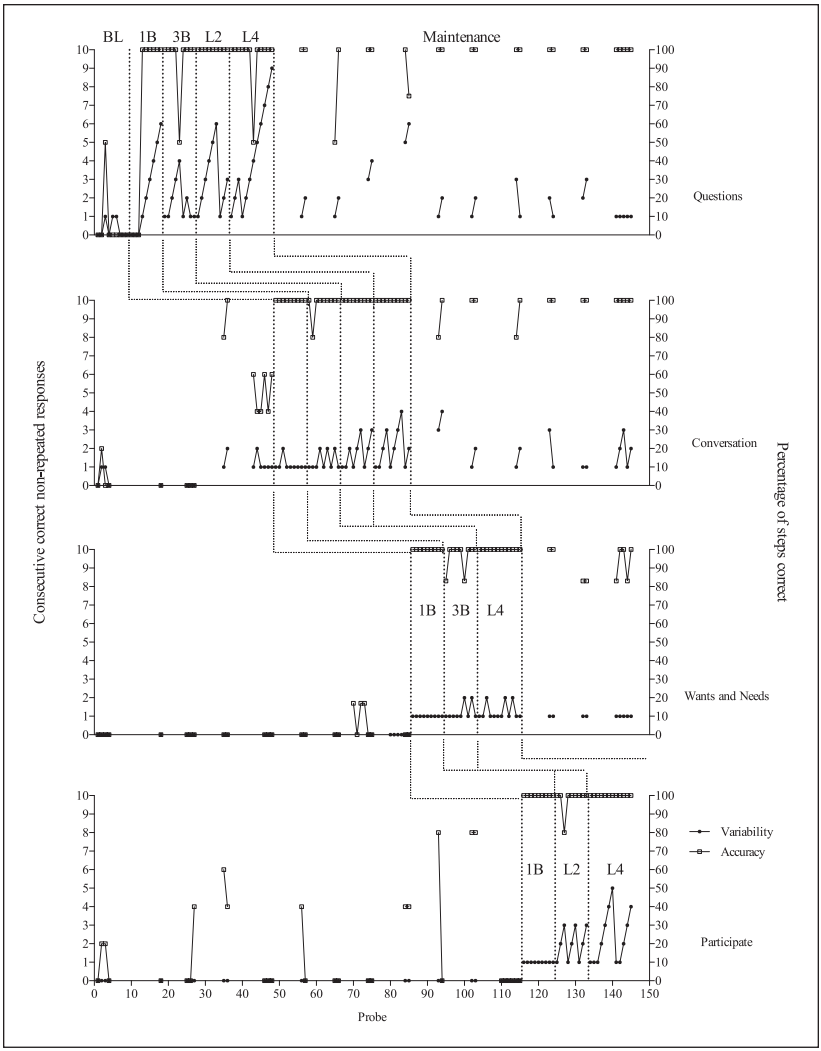


Figure 5. Consecutive correct non-repeated responses and percentage of steps correct, Tony.

correct non-repeated responses for Responding to Questions, similar to the two preceding experimental phases. For Maintaining a Conversation and Participate, introduction of Lag 2 was associated with a greater number of consecutive correct non-repeated responses than all previous phases. Non-overlap

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with previous phases was observed for Maintaining a Conversation and Participate. Due to absence, Lag 2 data were not collected for Expressing Wants and Needs. For all skills, introduction of Lag 4 was associated with the greatest number of consecutive correct non-repeated responses. However, data for Expressing Wants and Needs demonstrated much less appropriate variability than data for other skills. In addition, all skills except Expressing Wants and Needs demonstrated non-overlap with previous phases during Lag 4. Finally, variable responding somewhat maintained for Responding to Questions and Maintaining a Conversation, whereas it was not maintained for Expressing Wants and Needs.

During baseline measures, Tony demonstrated variable, low-to-moderate levels of skill accuracy across skills. An immediate increase percentage of steps completed correctly was observed during training with one exemplar, with minimal overlap with baseline data points and reduced appropriate variability. Similar levels of skill accuracy, appropriate variability, and overlap were observed across subsequent experimental phases. Tony's data indicate that improved accuracy of skills was maintained following termination of direct intervention.

Social Validity

The BIRS was completed by each participant's parent to evaluate the social validity of the intervention. Mark's mother reported overall acceptability (M = 4.5), indicating that she found the intervention to be acceptable (M = 4.9), slightly effective (M = 3.4), and had fast time to effectiveness (M = 5.0). Jarvis's mother reported similar overall acceptability (M = 4.9), indicating that she found the intervention to be very acceptable (M = 5.5), effective (M = 4.1), and did not have a fast time to effectiveness (M = 3.0). Phillip's mother indicated she found the intervention to be very acceptable (M = 5.9), very effective (M = 5.7), and had very fast time to effectiveness (M = 6.0), resulting in high overall acceptability (M = 5.5). Jarrod's mother responses indicated high overall acceptability (M = 5.2), reporting the intervention to be acceptable (M = 5.3), effective (M = 5.1), and had a fast time to effectiveness (M = 4.0). Finally, Tony's mother indicated high overall acceptability (M = 5.0), that she found the intervention to be acceptable (M = 4.9), effective (M = 5.0), and had fast time to effectiveness (M = 5.0).

Discussion

The purpose of the present study was to add to the literature supporting lag schedules of reinforcement for use in promoting appropriate variability.

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Specifically, this study sought to increase the variability of appropriate verbalizations in four children with ASD in response to four cues corresponding to four target social skills contained in the Superheroes Social Skills intervention program (Jenson et al., 2011). Traditionally, promoting appropriate variability in responding in social skills training programs has been done through the training of multiple exemplars (e.g., Plavnick et al., 2015). The results of this study demonstrated that multiple exemplar training did not substantially improve appropriate variability consistently across skills or participants; however, once the Lag 2 and Lag 4 schedules were introduced, the number of consecutive correct non-repeated responses increased for nearly all participants across all skills. In fact, for many participants, the number of cumulative correct novel responses greatly exceeded the criterion (e.g., 10 for Jarrod's responses to the Conversation skill) established for the lag schedule of reinforcement. These results support the previous literature demonstrating the use of lag schedules specifically for this purpose (Susa & Schlinger, 2012; Wolfe et al., 2014). Furthermore, there was no indication that participants were engaging in higher order stereotypy. Unfortunately, maintenance of the increased appropriate variability in responding was not consistent with most participants demonstrating a return to near baseline levels.

Despite the fact that the frequency of varied responding and degree of non-overlap was typically greatest during Lag 4, it must be noted that increases in variable responding were not similar across skills. Particularly, improvements in the frequency of appropriate variability and non-overlap during lag phases was often small for the Expressing Wants and Needs and Participate skills in comparison with other skills taught. One explanation for this finding may be that fewer correct and novel responses exist for the prompts use to probe Expressing Wants and Needs and Participate. For example, there are fewer possible ways to give a signal that you want to talk as a step of Expressing Wants and Needs (e.g., tap on shoulder, say name, say "excuse me") than demonstrating the final step of Responding to Questions (e.g., "tell me about your day"). Similarly, there are likely fewer appropriate responses to Step 5 (i.e., join in) of Participate than responses to Step 5 (i.e., make an appropriate comment) of Maintaining a Conversation. As such, smaller improvements in appropriate variability may, in part, be an artifact of the responses being measured. Alternatively, it must also be noted that appropriate levels of response appropriate variability have yet to be identified. As such, it is unknown whether the levels of variability observed for the Expressing Wants and Needs and Participate skills represent typical levels of appropriate variability, despite being smaller improvements over baseline levels.

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On the contrary, the secondary dependent variable, skill accuracy, showed nearly uniform improvements across participants and skills well above baseline levels. These improvements were initially observed during the initial intervention phase consisting of one trained exemplar and a FR schedule of reinforcement but persisted throughout all of the following intervention phases. This is likely due to the fact that the participants received rigorous training on skill accuracy immediately prior to the first intervention phase and no such training was provided for appropriate variability. In addition, skill accuracy maintained well above baseline levels for all participants across most skills during the maintenance phase albeit with greater variability.

Finally, the parents of each participant rated the intervention procedures as socially valid. Specifically, the procedures were rated as acceptable and effective by all parents. All parents except for one (Jarvis's mother) indicated that the intervention procedures also produced behavior change in a timely manner. These data add to the literature supporting the Superheroes Social Skills program as socially valid (e.g., Radley, Jenson, Clark, & O'Neill, 2014) and suggest that parents of the participants viewed the results as meaningful.

Limitations

To put the results of this study in the proper context, there are several limitations that should be addressed. First, there was no data collected on the generalization across stimuli, persons, or settings of either dependent variable. That is, it is unknown whether the participants demonstrated context appropriate or variable responses or exhibited accurate social skills in other contexts. This is particularly important given the difficulty of promoting generalization in social skills training for children with ASD (Rao, Beidel, & Murray, 2008). Although there is evidence to support the generalization of social skills using the Superheroes Social Skills package (e.g., Radley, O'Handley, et al., 2014), the lack of generalization in this study prevent similar conclusions regarding appropriate variability. It is, however, important to note that participants demonstrated response generalization by generating novel responses beyond those trained during the Lag 4 phase. Although this finding indicates that lag schedules may be useful for promoting response generalization, future research should program for the collection of generalization probes in other settings (e.g., classroom, community) and in the presence of individuals not related to the study (e.g., parents, novel peers).

Second, it is entirely possible that the appropriate variability in responding observed under the lag schedules of reinforcement were actually under the control of the rules provided to the participants during behavioral skills

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training. That is, participants were informed that they would be required to provide a specific number of novel responses before a mark on the self-monitoring card would be provided in the two lag schedule phases (i.e., Lag 2 and Lag 4). Thus, the participants' appropriate variability may have been rule-based instead of contingency-shaped. Due to the design of this study, it is impossible to separate the effects that the presentation of the rules and the lag schedule of reinforcement had on appropriate variability. However, it must be noted that the rules were only provided during behavioral skills training and not prior to probes of skill accuracy and appropriate variability. Future research should attempt to isolate these effects in this specific context, as previous research has done in more basic settings (e.g., Hayes, Brownstein, Zettle, Rosenfarb, & Korn, 1986; Newman, Buffington, & Hemmes, 1991).

Implications

Even in light of the limitations, the results hold important implications. First, because lag schedules of reinforcement appear to promote greater appropriate variability in verbal responses over FR schedules, practitioners may consider using them when appropriate variability is the primary goal of social skills training. For practitioners who may be unfamiliar with using lag schedules of reinforcement for this purpose, it may be easiest to conceptualize them as a FR schedule of reinforcement where novelty in verbal responses is required to satisfy the contingency. Second, although it is unclear whether the appropriate variability in responding was driven by rule-governed behavior or contingency-shaped behavior, practitioners wanting to use lag schedules in this manner should continue to provide children with rules about how the lag schedules operate until future research addresses this question. Finally, and perhaps most importantly, results of the study demonstrate that lag schedules of reinforcement resulted in response generalization. In other words, participants demonstrated novel, untrained responses when lag schedules were in place, resulting in a greater variety of correct responses than what was required to satisfy the criterion for reinforcement under the lag schedule. As such, the current study provides initial evidence regarding the potential utilization of lag schedules of reinforcement as an efficient strategy for increasing the number of correct responses in an individual's repertoire without training all possible responses.

Conclusion

In conclusion, the results of this study demonstrate the utility of lag schedules of reinforcement to promote appropriate variability in children with ASD.

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Although the results are promising, more research in this area is warranted (e.g., need for provision of rule regarding number of response exemplars required to meet lag schedule requirements) before widespread adoption of this practice into social skills training is recommended.

Declaration of Conflicting Interests

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